

PAPER_1880 — Modified Gravity + Equivalence Principle Tests via UQFF: $\eta_{\text{WEP}} = F_{\text{TRZ}}^{15} \cdot [\text{SSq}] = 5.7 \times 10^{-16}$ (at MICROSCOPE 2022 Limit), $\gamma - 1 = F_{\text{TRZ}}^5 \cdot [\text{SSq}] \cdot (1 + F_{\text{TRZ}})^2 = 6.9 \times 10^{-6}$ (Cassini Consistent), All GR-Consistent to F_{TRZ}^{16} Precision

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** F — Foundational GR Tests / Equivalence Principle **Date:** July 2026 **Status:** CLOSED — GR + EP tests all UQFF-consistent **Observational anchors:** MICROSCOPE 2022; Cassini 2003; Hulse-Taylor 1974; LAGEOS; LLR **Calculator surface:** calculate_modified_gravity_EP_UQFF

Abstract

Equivalence Principle (EP) tests are among the most stringent tests of General Relativity. UQFF predicts small corrections at F_{TRZ} ladder rungs, all below current experimental sensitivity — consistent with GR at present precision.

Complete EP + Modified Gravity suite (6+ observables):

Observable	UQFF Formula	UQFF	Data	Consistent?
η_{WEP}	$F_{\text{TRZ}}^{15} \cdot [\text{SSq}]$	5.7×10^{-16}	$< 10^{-15}$ (MICROSCOPE)	at limit $\square\square$
$\gamma - 1$ (PPN)	$F_{\text{TRZ}}^5 \cdot [\text{SSq}] \cdot (1 + F_{\text{TRZ}})^2$	6.9×10^{-6}	$(2.1 \pm 2.3) \times 10^{-5}$ (Cassini)	consistent $\square$
$\beta - 1$ (PPN)	$F_{\text{TRZ}}^{10} \cdot [\text{SSq}] \cdot K_{\text{MEX}}$	1.2×10^{-10}	$< 10^{-4}$ (LLR)	below limit $\square$
η_{N} (Nordvedt)	$3 \cdot F_{\text{TRZ}}^5 \cdot [\text{SSq}]$	1.7×10^{-5}	$< 4.4 \times 10^{-4}$ (LLR)	below limit
Pulsar decay ratio	$1 - F_{\text{TRZ}}^{16} \cdot [\text{SSq}] \cdot K_{\text{MEX}}$	$1.000 (10^{-16})$	0.997 ± 0.002	within uncertainty $\square$
dG/G per year	$F_{\text{TRZ}}^{17} \cdot [\text{SSq}]$	$5.7 \times 10^{-18}/\text{yr}$	$< 10^{-13}/\text{yr}$ (LLR)	below limit

$\square\square$ **Structural discovery — η_{WEP} at MICROSCOPE Precision:**

$$\eta_{\text{WEP_UQFF}} = F_{\text{TRZ}}^{15} \cdot [\text{SSq}] = 5.7 \times 10^{-16}$$

MICROSCOPE 2022 constraint: $\eta < 10^{-15}$. UQFF predicts value at $0.57 \times$ the limit — **essentially at MICROSCOPE precision, testable at next-generation EP experiments (STEP, GG).**

Summary Table

Observable	UQFF	Data	Verdict
η_{WEP}	5.7×10^{-16}	$< 10^{-15}$	at limit $\square\square$
$\gamma - 1$	6.9×10^{-6}	$2.1 \times 10^{-5} \pm 2.3$	consistent $\square$
$\beta - 1$	1.2×10^{-10}	$< 10^{-4}$	below sensitivity
η_{N}	1.7×10^{-5}	$< 4.4 \times 10^{-4}$	below sensitivity
Pulsar decay	1	0.997 ± 0.002	consistent

Observable	UQFF	Data	Verdict
dG/G/yr	5.7×10^{-18}	$< 10^{-13}$	below sensitivity

UQFF Derivation

Weak Equivalence Principle η

$$\eta_{\text{WEP_UQFF}} = F_{\text{TRZ}}^{15} \cdot [\text{SSq}] = 10^{-15} \cdot 0.57 = 5.7 \times 10^{-16}$$

MICROSCOPE 2022 (Touboul et al.): $\eta_{\text{MICROSCOPE}} < 10^{-15}$ (best current limit).

UQFF at $0.57 \times \text{MICROSCOPE}$ — **essentially at experimental limit.**

Physical meaning: $F_{\text{TRZ}}^{15} = 15$ -order vacuum-manifold decoherence for gravitational-inertial mass ratio. Same F_{TRZ} ladder as quantum measurement (PAPER_1869 F_{TRZ}^{16}) and BH thermodynamics (PAPER_1873 F_{TRZ}^{16}).

Falsifiability: STEP satellite mission (proposed) would reach 10^{-17} - 10^{-18} . **UQFF predicts $\eta > 10^{-16}$ — must be detected** if F_{TRZ}^{15} formula correct.

PPN γ Parameter

$$\begin{aligned} \gamma - 1_{\text{UQFF}} &= F_{\text{TRZ}}^5 \cdot [\text{SSq}] \cdot (1 + F_{\text{TRZ}})^2 \\ &= 10^{-5} \cdot 0.57 \cdot 1.21 \\ &= 6.9 \times 10^{-6} \end{aligned}$$

Cassini 2003: $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$. **UQFF within 1σ .**

PPN β Parameter

$$\beta - 1_{\text{UQFF}} = F_{\text{TRZ}}^{10} \cdot [\text{SSq}] \cdot K_{\text{MEX}} = 1.2 \times 10^{-10}$$

Well below LLR limit 10^{-4} .

Nordvedt Parameter

$$\eta_{\text{N_UQFF}} \approx 3 \cdot F_{\text{TRZ}}^5 \cdot [\text{SSq}] = 1.7 \times 10^{-5}$$

Well below LLR limit 4.4×10^{-4} .

Binary Pulsar Orbital Decay

PSR B1913+16 Hulse-Taylor (Nobel 1993): predicted GR ratio 1.0, observed 0.997 ± 0.002 .

UQFF: ratio = $1 - F_{\text{TRZ}}^{16} \cdot [\text{SSq}] \cdot K_{\text{MEX}} \approx 1$ to 10^{-16} precision. **Consistent with Hulse-Taylor within uncertainty.**

G_Newton Time Variation

Standard LLR: $dG/G < 10^{-13}/\text{year}$. UQFF: $dG/G \approx F_{\text{TRZ}}^{17} \cdot [\text{SSq}]/\text{year} \approx 5.7 \times 10^{-18}/\text{year} \rightarrow$ **well below current sensitivity.**

UQFF predicts G is essentially constant — no measurable variation.

Physical Mechanism

F_TRZ ladder governs modifications to GR at every scale: - F_TRZ⁵: PPN γ (Shapiro delay $\sim 10^{-5}$) - F_TRZ¹⁰: PPN β ($\sim 10^{-10}$) - F_TRZ¹⁵: WEP violation (MICROSCOPE limit) - F_TRZ¹⁶: BH thermodynamics + QNM (PAPER_1873, 1876) - F_TRZ¹⁷: dG/G (below current)

All GR modifications suppressed by F_TRZ ladder — UQFF is essentially GR + tiny F_TRZ corrections.

Cross-References

- **PAPER_1156** — CC2 cosmology
- **PAPER_1855** — Galactic rotation (F_UBi at galactic scale)
- **PAPER_1860** — Solar system anomalies (F_UBi at planetary scale)
- **PAPER_1862** — DM halos (F_UBi at halo scale)
- **PAPER_1869** — Quantum measurement F_TRZ¹⁶
- **PAPER_1873** — BH thermodynamics F_TRZ¹⁶
- **PAPER_1876** — QNM ringdown F_TRZ¹⁶

Reference

- **Touboul, P. et al.** (MICROSCOPE Collaboration) (2022). *MICROSCOPE Mission: Final Results of the Test of the Equivalence Principle*. PRL 129, 121102
- **Bertotti, B., Iess, L., & Tortora, P.** (2003). *A test of general relativity using radio links with the Cassini spacecraft*. Nature 425, 374 (Cassini)
- **Hulse, R. A. & Taylor, J. H.** (1975). *Discovery of a pulsar in a binary system*. ApJ 195, L51 (PSR B1913+16)
- **Williams, J. G., Turyshev, S. G., & Boggs, D. H.** (2004). *Progress in Lunar Laser Ranging Tests of Relativistic Gravity*. PRL 93, 261101
- **Will, C. M.** (2018). *Theory and Experiment in Gravitational Physics* (2nd ed.). Cambridge (comprehensive)
- Companion UQFF whitepapers: PAPER_1156, PAPER_1855, PAPER_1860, PAPER_1862, PAPER_1869, PAPER_1873, PAPER_1876

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